

Standing Waves

We saw in the previous note that the wave equation tells us that the motion of a string under tension looks like

$$y = f(x \pm vt) \tag{1}$$

where y is the height of the string, x is the position along the string, and t is the time at which we look at the string. The velocity v is the speed at which waves move along the string, given by $\sqrt{\frac{T}{\lambda}}$ where T is the tension (force) in the string, and λ is the mass-per-unit-length (so if 3 metre string had a mass of 6 kg, we would have $\lambda = 6\text{kg}/3\text{m} = 2\text{kg/m}$).

The function f can be *any* function. The way I would think about this is to picture the spring in class. However I jiggled my hand, the same pattern moved down the string. The function f corresponds to how I moved my hand, and I am always free to move my hand however I like. However, once I do move my hand, then the wave equation tells us that the pattern of how I moved my hand will travel down the string with velocity v . The velocity can be either positive or negative, so signals can move either to the right or the left, but every signal has to travel at the same speed. No matter how I jiggle my string, the signal will move away at speed v .

It's also important to know that if I have two different solutions to the wave equation, then the sum of those two solutions is *also* a solution. That means if I have two functions $f(x - vt)$ and $g(x + vt)$ that solve the wave equation, then $f(x - vt) + g(x + vt)$ *also* solves the equation. In this case, f is moving to the right and g is moving to the left. The sum of them $f + g$ also solves the wave equation. The way I like to think about this is I have a very long string with a person at either end. The person on the left shakes the left end of the string with function f and the person on the right shakes their end with function g . When the two waves hit in the middle of the string, the height of the string is just the sum of the two heights. You can look [here](#) to see a positive wave from the left hitting a negative wave from the right. The red dashed line is the right-moving wave, the green dashed line is the left-moving wave, and blue is the sum.

1. Waves Hitting a Boundary

We're working on making our way towards how a string behaves, but we aren't there yet. The next step on the road is to look at what happens if I nail down one end of the string and shake the other end. As before, f describes how I shake my end of the string. Now, though, the other end of the string has to always have height $y = 0$. Why? Well, because we nailed it down so that end can't move. If I just took my function f then this can no longer be the whole story because when the wave hits the end of the string, it would move.

What we have to do is add *another* wave that keeps the end of the string fixed. So now we pretend the string is much longer but is pinned to zero at a point in the middle. We send our wave down, and the person at the other end of the string sends a very carefully timed wave as well. They time it so that the two waves cancel out exactly at the point where we pinned down the string. Now we have something that is a solution to the wave equation and *also* has $y = 0$ where we pinned

the string. The wave our friend at the other end to send is exactly the wave we sent, but flipped around and upside down. If we send a positive bump, our friend has to send a negative bump.

If you go back and look at the desmos link, you can see that's exactly what's happening. A positive bump comes in from the right, and a negative from the left. I've timed them so they hit $x = 0$ at the same time, and so you can see the height doesn't move there. The final step is to say "Hey, we have a solution that keeps the end of our original string fixed. Do we actually need the (imaginary) rest of the string?" The answer is an emphatic "No!" Because we found a solution that matches the string we have, we're done. When we a wave to the end of a fixed string, we get the same wave back, but flipped around. The wave *reflected* off the end of the string. You probably could have guessed that some sort of reflection would happen, but now we know exactly what that reflection looked like: a flipped copy of the original signal.

2. Waves hitting Two Boundaries

We're getting close, but we don't have a violin (or piano, or cello, or harp...) just yet. An instrument has the string tied down at *both* ends. We're going to play the same game, but we'll see that having to match a wave at two points makes a huge difference. First, picture a bump coming in from the left, but now we have our total signal be zero at two points, say $x = \pm 2$. Well, it's easy enough to manage our incoming bump. We just send two negative bumps from the right, one for each point where we need to be zero. [Look here](#) to see this in action. We have one bump moving from the left, and we want that bump to be cancelled at both $x = +2$ and $x = -2$. We send our two bumps in from the right, and presto, the left bump is cancelled both times. We're done, right? Unfortunately no, because if you go back and watch the bumps coming in from the right, you can see when the first bump hits $x = +2$, there's nothing there to cancel it. Similarly, when the second bump hits $x = -2$, it's also not cancelled. The string is nailed down, so whatever solution we have must *always* be zero at these points. "Not a problem!" I hear you say. "We'll send more bumps from the right!" Adding a second bump [looks like this](#). Now our string at $x = -2$ stays at zero, but the second bump isn't cancelled at $x = +2$.

If we have some number of bumps coming from the right, we need at least one more bump from the left to cancel at two points. Whenever we have some number of bumps from the left, we need at least one more bump from the right. The only way this can be true is if we have an *infinite* number of bumps coming from both directions. The easisest way to send an infinite number of bumps is a sine wave, so let's look at a pair of opposite sinewaves. My desmos plot is [here](#). I suggest you look at this for a while and look at the individual pieces. The right-moving sine is red, the left-moving is blue, and the sum of the two waves is black. The combination of these two waves now no longer looks like a signal travelling along a string. Instead, we see a stationary pattern, with every point just going up and down. There are now not just two points, but a whole bunch of points that don't move at all. These non-moving (or stationary) points are called *nodes*, and we could have a string going between any pair of nodes.

3. Standing Waves

Mathematically, let's start with a pair of sine waves $\sin(kx - \omega t)$ and $\sin(kx + \omega t)$. We saw in class that the speed at which a wave moves is $v = \omega/k$, so if we pick k then we know what ω is, or if we pick ω , we know what k is. If we add these two waves together, we get

$$\sin(kx - \omega t) + \sin(kx + \omega t) = \sin(kx) \cos(\omega t) - \cos(kx) \sin(\omega t) + \sin(kx) \cos(\omega t) + \cos(kx) \sin(\omega t) \quad (2)$$

$$= 2 \sin(kx) \cos(\omega t) \quad (3)$$

If you look at Equation 3, you can see that if we pick a value of x , then the height goes between $\pm 2 \sin(kx)$ with angular frequency ω . The amplitude goes to zero wherever the $\sin(kx)$ goes to zero. These are called *standing waves* because the string is going up and down, but the signals don't look like they're moving left or right. Recall that $\sin(0, \pi, 2\pi, 3\pi \dots) = \sin(-\pi, -2\pi, -3\pi \dots) = 0$. More generally, we know that $\sin(n\pi) = 0$ whenever n is an integer.

Now we want to find out what values of k we're allowed to have. If we have a very long string with no constraints, we could pick any value of k . However, if we have a physical string, then we'll see that only certain value of k are allowed. Let's say we have a string that goes from 0 to π , and is nailed down at the ends. We know that $\sin(kx)$ must be zero whenever $x = 0$ or $x = \pi$. The $x = 0$ point is easy, because $\sin(0) = 0$. However, $x = \pi$ is a little more complicated. We know that kx must be equal to $n\pi$ for some integer n when $x = \pi$. We can plug in $x = \pi$ now, and get

$$k\pi = n\pi \quad (4)$$

or

$$k = n \quad (5)$$

Remember, we need that n is an integer, so Equation 5 tells us that $\sin(kx) = 0$ at $x = \pi$ as long as k is an integer. You can play with this at [this plot](#). If you pick an integer for k , then the sine is always 0 at $x = 0$ and $x = \pi$. If you pick a non-integer for k , then the sine does not go to zero. If I tell you that the string is pegged at $x = 0$ and $x = \pi$, then we know our sine solution has to look like $\sin(kx)$ where k can be *any* integer.

4. Frequency of Oscillations

We basically have an instrument now. Let's look at how quickly the sines oscillate. We know that our solution looks like $\sin(kx) \cos(\omega t)$ where we know that k is an integer (we're still keeping our string π long). We know that $v = \omega/k$, or $\omega = kv$. The speed v is constant (the wave equation told us that), so now if k has to be an integer, then the *only* allowed values for ω are some integer times v . The higher the value of k , the higher the value of ω , and the faster the string moves up and down. If we tie a string down at both ends and put some tension on the string, then the string is going to oscillate at a finite set of frequencies, and that these frequencies are integer multiples of each other.

I haven't shown it mathematically, but it will turn out that *any* motion of the string can be described as the sum of these sine waves. The slowest the string can oscillate is when $k = 1$ and

the string goes up and down with angular frequency ω . Low frequency means low pitch, so the lowest note we could ever play on our string has angular frequency ω . We could also play a note with frequency 2ω or 3ω and so on. You can't really do this with a violin, but we'll see that brass instruments behave the same way. A bugle has no valves at all, and when someone plays different notes with a bugle, essentially what they're doing is switching between the different allowed values of k and ω . That's why you can play *taps* with a bugle, but not a scale. There's a lowest note we could possibly play, and that note is called the *fundamental*. The higher notes are called *overtones* or *harmonics*.

5. Wavelengths, Frequencies, and String Lengths

We have had lots of variables tossed around which can get rather confusing. I'll go through these terms now and establish their relations. Once we're a bit more comfortable with these, I'll switch to an arbitrary string length, since not all strings are going to be exactly π metres long.

The first thing to nail down is the concept of a *wavelength*. If I have a sine/cosine wave, a wavelength is the length it takes for a sine to go around a full circle/through an entire cycle of 2π radians. If I have $\sin(x)$ then I get a full wavelength then x goes from 0 to 2π , or 2π to 4π , or π to 3π etc. If we call the wavelength λ then we have $\lambda = 2\pi$ for $\sin(x)$. What about $\sin(kx)$? The wavelength will be the value of x when $kx = 2\pi$. We have $x = 2\pi/k = \lambda$, or $k = 2\pi/\lambda$. We could replace k with $2\pi/\lambda$ to get

$$\sin\left(\frac{2\pi}{\lambda}x\right) \quad (6)$$

For our string of length π , we know the allowed solutions look like $\sin(kx)$. If $k = 1$, then the wavelength is $\lambda = 2\pi/k = 2\pi$. The fundamental of a string of length π has a wavelength not of π , but of 2π . The wavelength of the fundamental is twice as long as the string we have, so the string can oscillate with half a wavelength. We'll see later on that if we set up our instrument differently, we can get the fundamental wavelength to be four times longer. That will have a significant impact on how the instruments sound to our ear.

In music, the most common way of talking about the time part of a wave is *frequency*, denoted ν . The analog to wavelength (the distance it takes a wave to repeat) is the *period*, denoted T , which is the length of time it takes a wave to repeat. If I have a period of 0.1 seconds, then the wave repeats $1/0.1 = 10$ times per second. That gives the relation between ν and T :

$$\nu = 1/T \quad (7)$$

I know we have a full wavelength of $\sin(\omega t)$ when $\omega t = 2\pi$. That happens when $t = T$, that is, the time is equal to one full period. That gives $\omega T = 2\pi$ or

$$T = 2\pi/\omega \quad (8)$$

Combining Equations 7 and 8 we have $2\pi/\omega = 1/\nu$ or

$$\omega = 2\pi\nu \quad (9)$$

Equation 9 tells us that frequency and ω just differ by a factor of 2π . This isn't really a surprise - ω is the rate at which we move by a radian, and ν is the rate at which we move by 2π radians. Since we have to go 2π times further for ν , the rate is going to be $\omega/2\pi$. We normally call ν the frequency, and ω the *angular frequency*.

Finally, let's look at what happens when we have a string of length L . We know our solutions still look like $\sin(kx)\sin(\omega t)$, but now we need our $\sin(kx)$ to go to zero whenever $x = L$. That means $kL = n\pi$ or

$$k = \frac{n\pi}{L} \quad (10)$$

We still know that $v = \omega/k$, so the allowed values of ω are where $\omega = vk$ or:

$$\omega = \frac{n\pi v}{L} \quad (11)$$

Finally, since frequency $\nu = \omega/2\pi$, we have the allowed frequencies:

$$\nu = \omega/2\pi = \frac{n\pi v}{2\pi L} = \frac{nv}{2L} \quad (12)$$

This is an important result if you ever want to build an instrument! If you take a string of length L and the waves move at speed v , then the frequencies you ear will hear are given by Equation ???. The lowest frequency we'll ever get is where $n = 1$, so our fundamental frequency will be $\frac{v}{2L}$.

Let's work our the wave speed on a violin as an example. When you have a violin, you know what frequencies you want to have in order to match the rest of the orchestra. The A string should be at 440 Hz , and the length L is about 0.32m between where the string is held fixed. We know then that $440\text{ Hz} = \frac{v}{2 \times 0.32\text{m}}$ or $V = 440 \times 2 \times 0.32\text{m} = 281.6\text{m/s}$. When you tune a violin, you are adjusting the tension in the string until the velocity matches this value.